

Cryptology Assignment - 3

1. Determine the multiplicative inverse modulo 26 for each of the following integers:

$\rightarrow 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25 \pmod{26}$.

Ans: (i) Inverse of 1.

$$\Rightarrow 1^{-1} \pmod{26} = 1$$

$$26 = 1(1)$$

(ii) $3^{-1} \pmod{26}$

$$\Rightarrow 26 = 3(8) + 2 \quad 2 = 26 + 3(-8)$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$3(9) + 26(-1) = 1$$

$$\text{Inverse} = \underline{\underline{9}}$$

(iii) $5^{-1} \pmod{26}$

$$\Rightarrow 26 = 5(5) + 1$$

$$1 = 26 + 5(-5)$$

$$= 26(1) + 5(-5)$$

$$= -5 \pmod{26}$$

$$\text{Inverse} \Rightarrow \underline{\underline{21}}$$

(iv) $7^{-1} \pmod{26}$

$$\Rightarrow 26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 = 26 + 7(-3)$$

$$2 = 7 + 5(-1)$$

$$5 + 2(-2) = 1$$

$$[26 + 7(-3)] + [7 + 5(-1)](-2) = 1$$

$$[26 + 7(-3)] + [7 + [26 + 7(-3)](-1)](-2) = 1$$

$$[26 + 7(-3)] + [7 + [26 + 7(-3)](-1)](-2) = 1$$

$$[26(1) + 7(-3)] + [7 + [26(-1) + 7(3)]](-2) = 1$$

$$[26(1) + 7(-3) + [7(1) + [26(-1) + 7(3)]](-2)](-2) = 1$$

$$\{26(1) + 7(-3) + 7(-2) + 26(2) + 7(-6)\} = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = \underline{\underline{15}}$$

$$\text{Inverse} = \underline{\underline{15}}$$

(v) $9^{-1} \bmod 26$

$$26 = 9(2) + 8$$

$$8 = 26 + 9(-2)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$9(1) + [26 + 9(-2)](-1) = 1$$

$$9(1) + 26(-1) + 9(2) = 1$$

$$9(3) + 26(-1) = 1$$

$$\text{Inverse} = \underline{\underline{3}}$$

(vi) $11^{-1} \bmod 26$

$$26 = 11(2) + 4$$

$$4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$4 + 3(-1) = 1$$

$$[26 + 11(-2)] + [11 + 4(-2)](-1) = 1$$

$$[26 + 11(-2)] + [11 + [26 + 11(-2)](-2)](-1) = 1$$

$$[26 + 11(-2)] + [11 + 26(-2) + 11(4)](-1) = 1$$

$$[26 + 11(-2)] + [11(-1) + 26(2) + 11(-4)] = 1$$

$$26(1) + 11(-2) + 11(-1) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\text{Inverse} = \underline{\underline{19}}$$

(vi) $15^{-1} \text{ mod } 26$

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$11 = 26 + 15(-1)$$

$$4 = 15 + 11(-1)$$

$$3 = 11 + 4(-2)$$

$$4 + 3(-1) = 1$$

$$[15 + 11(-1)] + [11 + 4(-2)](-1) = 1$$

$$[15 + [26 + 15(-1)](-1)] + [[26 + 15(-1)] + [15 + 11(-1)](-2)](-1) = 1$$

$$[15 + [26 + 15(-1)](-1)] + [[26 + 15(-1)] + [15 + [26 + 15(-1)](-1)](-2)](-1) = 1$$

$$[15 + [26(-1) + 15(1)]] + [[26 + 15(-1)] + [15 + [26 + 15(-1)](-1)](-2)](-1) = 1$$

$$15 + [26(-1) + 15(1)] + [[26 + 15(-1)] + [15 + 26(-1) + 15(1)](-2)](-1) = 1$$

$$15(1) + 26(-1) + 15(1) + 26(-1) + 15(1) + 15(2) + 26(-1) + 15(2) = 1$$

$$\text{Inverse} = \underline{\underline{7}}$$

(vii) $17^{-1} \text{ mod } 26$

$$26 = 17(1) + 9$$

$$9 = 26 + 17(-1)$$

$$17 = 9(1) + 8$$

$$8 = 17 + 9(-1)$$

$$9 = 8(1) + 1$$

$$9 + 8(-1) = 1$$

$$[26 + 17(-1)] + [17 + 9(-1)](-1) = 1$$

$$[26 + 17(-1)] + [17 + [26 + 17(-1)](-1)](-1) = 1$$

$$[26 + 17(-1)] + [17 + 26(-1) + 17(1)](-1) = 1$$

$$26(1) + 17(-1) + 17(-1) + 26(1) + 17(-1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26 - 3 = \underline{\underline{23}}$$

$$\text{Inverse} = \underline{\underline{23}}$$

ix $19^{-1} \bmod 26$

$$26 = 19(1) + 7$$

$$7 = 26 + 19(-1)$$

$$19 = 7(2) + 5$$

$$5 = 19 + 7(-2)$$

$$7 = 5(1) + 2$$

$$2 = 7 + 5(-1)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[19 + 7(-2)] + [7 + 5(-1)](-2) = 1$$

$$[19 + [26 + 19(-1)](-2)] + [[26 + 19(-1)] + [19 + 7(-2)](-1)](-2) = 1$$

$$[19 + [26 + 19(-1)](-2)] + [[26 + 19(-1) + [19 + [26 + 19(-1)](-2)](-1)](-2) = 1$$

$$19 + 26(-2) + 19(2) + [[26 + 19(-1) + [19 + 26(-2) + 19(2)](-1)](-2) = 1$$

$$19 + 26(-2) + 19(2) + [26 + 19(-1) + 19(-1) + 26(2) + 19(-2)](-2) = 1$$

$$19(1) + 26(-2) + 19(2) + 26(-2) + 19(2) + 19(2) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$\text{Inverse} = \underline{\underline{11}}$$

x $21^{-1} \bmod 26$

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$21 = 5(4) + 1$$

$$21 + 5(-4) = 1$$

$$21 + [26 + 21(-1)](-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$21(1) + 26(-4) + 21(4) = 1$$

$$21(5) + 26(-4) = 1$$

$$= 5$$

$$\text{Inverse} = \underline{\underline{5}}$$

xi $23^{-1} \bmod 26$

$$3 = 26 + 23(-1)$$

$$26 = 23(1) + 3$$

$$2 = 23 + 3(-7)$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$3 + 2(-1) = 1$$

$$[26 + 23(-1)] + [23 + 3(-7)](-1) = 1$$

$$[26 + 23(-1)] + [23 + [26 + 23(-1)](-7)](-1) = 1$$

$$[26 + 23(-1)] + [23 + 26(-7) + 23(7)](-1) = 1$$

$$26(1) + 23(-1) + 23(-1) + 26(7) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$\text{Inverse} = \underline{\underline{17}}$$

ii) $25^{-1} \bmod 26$

$$26 = 25(1) + 1$$

$$25 + 25(-1) = 1$$

$$26(1) + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\text{Inverse} = \underline{\underline{25}}$$

2. Choose any string of your choice to encrypt using the affine cipher with key values: multiplier 5 and additive 8.

Ans: Key values given: Multiplier $a=5$, additive $b=8$

Chosen plaintext: "HELLO"

Step 1:

The affine cipher must satisfy,

$$\gcd(a, 26) = 1$$

$$\text{So, } \gcd(5, 26) = 1$$

Hence, key is valid.

Step 2:

$$E(x) = (5x + 8) \bmod 26$$

where x is each letter's numeric value.

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Step 3: Converting the letters to numbers.

Letter	H	E	L	L	O
x	7	4	11	11	14

Step 4:

H ($x=7$):

$$(5 \times 7 + 8) \bmod 26 = (35 + 8) \bmod 26 \\ = 43 \bmod 26$$

$$43 = 1(26) + 17 \rightarrow 17 \Rightarrow R$$

E ($x=4$):

$$(5 \times 4 + 8) \bmod 26 = (20 + 8) \bmod 26 \\ = 28 \bmod 26$$

$$28 = 1(26) + 2 \rightarrow 2 \Rightarrow C$$

L ($x=11$):

$$(5 \times 11 + 8) \bmod 26 = (55 + 8) \bmod 26 \\ = 63 \bmod 26$$

$$63 = 2(26) + 11 \rightarrow 11 \Rightarrow L$$

L ($x=11$):

$$(5 \times 11 + 8) \bmod 26 = (55 + 8) \bmod 26 \\ = 63 \bmod 26$$

$$63 = 2(26) + 11 \rightarrow 11 \Rightarrow L$$

O ($x=14$):

$$(5 \times 14 + 8) \bmod 26 = (70 + 8) \bmod 26 \\ = 78 \bmod 26$$

$$78 = 3(26) + 0 \rightarrow 0 \Rightarrow A$$

$\therefore$ "HELLO" $\Rightarrow$ "RCLLA"

Step 6: Verify by decrypting back

Inverse of $5 \bmod 26 = 21$

Decryption formula: $D(y) = 21(y-8) \bmod 26$

So, let us check on 'R':

$$R(y=17): 21(17-8) \bmod 26 = 21(9) \bmod 26 \\ = 189 \bmod 26$$

$$189 = 7(26) + 7 \Rightarrow 7 \rightarrow \underline{H} \checkmark$$

Decrypting 'R' results in 'H'. Hence, the encryption is valid and reversible.

3. Encrypt 'HI' using the Affine Cipher with $a=11$ and $b=5$.

Ans: $\gcd(a, 26) = \gcd(11, 26) = 1$

Hence, key is valid.

$$E(x) = (11x + 5) \bmod 26$$

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

$$H \ I \Rightarrow H = 7 \quad I = 8$$

$$H(x=7) \Rightarrow (11 \times 7 + 5) \bmod 26 = (77 + 5) \bmod 26 \\ = 82 \bmod 26$$

$$82 = 3(26) + 4 \rightarrow 4 \Rightarrow E$$

$$I(x=8) \Rightarrow (11 \times 8 + 5) \bmod 26 = (88 + 5) \bmod 26 \\ = 93 \bmod 26$$

$$93 = 3(26) + 15 \rightarrow 15 \Rightarrow P$$

$$\therefore "HI" \Rightarrow "EP"$$

4. The ciphertext UCR was encrypted using the affine function $9x + 2 \bmod 26$. Find the plaintext.

Ans: To decrypt, we need a^{-1} , such that $9 \times a^{-1} \equiv 1 \pmod{26}$

By Euclidean algorithm:

$$26 = 2(9) + 8$$

$$9 = 1(8) + 1$$

$$8 = 8(1) + 0 \rightarrow \text{gcd} = 1$$

$$\cancel{1 \times 9}$$

$$8 = 26 + 9(-2)$$

$$9 + 8(-1) = 1$$

$$9(1) + 8(-1) = 1$$

$$9(1) + [26 + 9(-2)](-1) = 1$$

$$9(1) + 26(-1) + 9(2) = 1$$

$$26(-1) + 9(3) = 1$$

$$\text{Inverse} = \underline{3} = a^{-1}$$

$$D(y) = a^{-1}(y - b) \bmod 26$$

where y is each ciphertext letter's numeric value.

$$\frac{\text{letter}}{u} = \frac{y}{20}$$

$$c = 2$$

$$R = 17.$$

$$u(y) = u(20)$$

$$3(20 - 2) \bmod 26 = 3(18) \bmod 26$$
$$= 54 \bmod 26$$

$$= 2(26) + 2 \rightarrow 2 \Rightarrow \underline{C}$$

$$c(y = 2)$$

$$3(2 - 2) \bmod 26 = 0 \bmod 26$$

$$= 0 \Rightarrow \underline{A}$$

$$R(y = 17)$$

$$3(17 - 2) \bmod 26 = 3(15) \bmod 26$$

$$45 \bmod 26$$

$$45 = 1(26) + 19 \rightarrow 19 \Rightarrow \underline{T}$$

$\therefore \underline{\text{'UCR'}} \Rightarrow \underline{\text{'CAT'}}$ is the plaintext.

6. Decrypt the cipher text using Affine cipher, KDYXJUWDEXZY using key = 3 (Multiplier) and 4.

Ans

$$Decy = a^{-1}(y-b) \text{ mod } 26$$

Cipher text = KDYXJUWDEXZY

Multiplier, $a = 3$

key $b = 4$

$A = 0, B = 1, \dots, Z = 25.$

To find a^{-1} ,

$$3 \times a^{-1} \equiv 1 \pmod{26}$$

Since, $3 \times 9 = 27 \equiv 1 \pmod{26}$

$$\therefore 3^{-1} = 9$$

$$Decy = 9(y-4) \text{ mod } 26.$$

Cipher	Value	$9(y-4) \text{ mod } 26$	Plaintext
K	10	2	C
D	3	17	E
Y	24	24	Y
X	23	15	P
J	9	19	T
U	20	14	O
W	22	6	G
D	3	17	R
E	4	0	A
X	23	15	P
Z	25	7	H
Y	24	24	Y

$\therefore \text{KDYXJUWDEXZY} \Rightarrow \text{CRYPTOGRAPHY.}$

6. Use $k_1 = 13$ (multiplier), $k_2 = 4$ (Even though it is not a valid key - try to encrypt). Encrypt the string "input" and alter using the above key and find what is the issue.

Ans: Given:

$$k_1 = a = 13$$

$$k_2 = b = 4$$

Encryption: 'INPUT'

$$E(x) = (13x + 4) \bmod 26$$

$$I(x=8) \Rightarrow 108 \bmod 26 = 4 \Rightarrow \underline{\underline{E}}$$

$$N(x=13) \Rightarrow 173 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$$P(x=15) \Rightarrow 199 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$$U(x=20) \Rightarrow 264 \bmod 26 = 4 \Rightarrow \underline{\underline{E}}$$

$$T(x=19) \Rightarrow 251 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$\therefore \text{INPUT} \Rightarrow \text{ERRER}$

Encryption: 'ALTER'

$$E(x) = (13x + 4) \bmod 26$$

$$A(x=0) = 4 \Rightarrow \underline{\underline{E}}$$

$$L(x=11) = 147 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$$T(x=19) = 251 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$$E(x=4) = 56 \bmod 26 = 4 \Rightarrow \underline{\underline{E}}$$

$$R(x=17) = 225 \bmod 26 = 17 \Rightarrow \underline{\underline{R}}$$

$\therefore \text{ALTER} \Rightarrow \text{ERRER}$

The problem here is that 13 is not a valid multiplier for an Affine cipher. For an affine cipher, the multiplier has to be a coprime with 26. $\text{Gcd}(a, 26) = 1$.

But here,
 $\gcd(13, 26) = 13$.

Therefore, 13 has no multiplicative inverse modulo 26.
This causes different letters to encrypt to the same ciphertext letter.

The issue is that $k_1 = 13$ is an invalid affine cipher multiplier as $\gcd(13, 26) \neq 1$. Hence, the encryption is not reversible and multiple plaintext letters produce the same cipher text.

7. Find the inverse matrix of the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \pmod{26}$.

Ans: $K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \pmod{26}$.

$$K^{-1} = (\det K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \pmod{26}$$

$$= (3)(7) - (2)(5)$$

$$= 21 - 10 \Rightarrow \underline{\underline{11}}$$

Multiplicative inverse of 11 modulo 26.

$$11 \times 19 = 209$$

$$209 \pmod{26} = 1$$

$$\therefore 11^{-1} = 19 \pmod{26}$$

$$\text{adj}(K) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$\text{by taking modulo 26: } \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix} \pmod{26}$$

$$\Rightarrow \begin{bmatrix} 133 & 456 \\ 399 & 57 \end{bmatrix} \pmod{26}$$

$$\Rightarrow K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \pmod{26} \Rightarrow \begin{bmatrix} 3 & 14 \\ 5 & 7 \end{bmatrix}^{-1} \pmod{26} = \underline{\underline{\begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}}}$$

This is a valid key matrix as its determinant, 11 is coprime.

8. Find the inverse of the following matrix whose entries are considered modulo 26:

Ans: $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

For 2×2 matrix $K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant $ad - bc$.

$$\det(K) = (11)(3) - (13)(2) = 33 - 26 = \underline{7}$$

$$7^{-1} \pmod{26}.$$

$$\Rightarrow 26 = 3(7) + 5$$

$$5 = 26 + 3(-7)$$

$$7 = 1(5) + 2$$

$$2 = 7 + 1(-5)$$

$$5 = 2(2) + 1$$

$$5 + 2(-2) = 1$$

$$[26 + 3(-7)] + (7 + 1(-5))(-2) = 1$$

$$\Rightarrow \text{Inverse of } 7^{-1} \pmod{26} = \underline{15}.$$

$$\text{For } K = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ adj of } (K) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\text{adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

Convert negative entries to mod 26:

$$\bullet -13 \pmod{26} = 13$$

$$\bullet -2 \pmod{26} = 24$$

$$\text{adj}(K) \pmod{26} = \begin{bmatrix} 3 & 13 \\ 24 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \times \begin{bmatrix} 3 & 13 \\ 24 & 11 \end{bmatrix} \pmod{26}$$

$$(1,1): 15 \times 3 = 45 \rightarrow 45 - 26 = 19$$

$$(1,2): 15 \times 13 = 195 \rightarrow 195 = 7(26) + 13 \rightarrow 13$$

$$(2,1): 15 \times 24 = 360 \rightarrow 360 = 13(26) + 22 \rightarrow 22$$

$$(2, 2) \div 15 \times 11 = 165 \rightarrow 165 = 6(26) + 9 \rightarrow 9$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

9. Is the key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

Ans: $\det(K) = ad - bc = (2)(21) - (14)(13)$
 $= 42 - 182 = \underline{\underline{-140}}$

$$-140 \div 26 \Rightarrow -5.38 \dots \text{ so, } -6 \times 26 = \underline{\underline{-156}}$$

$$-140 - (-156) = -140 + 156 = 16$$

$$\det(K) \equiv 16 \pmod{26}$$

$$26 = 1(16) + 10$$

$$16 = 1(10) + 6$$

$$10 = 1(6) + 4$$

$$6 = 1(4) + 2$$

$$4 = 2(2) + 0$$

$$\gcd(16, 26) = 2$$

$\Rightarrow 26 = 2 \times 13$ and $16 = 2^4$ both share same factor.

Since, $\gcd(16, 26) = 2 \neq 1$, the determinant has no multiplicative inverse mod 26.

$\therefore$ This key is NOT VALID.

10. Encrypt using 2×2 Hill cipher for the plaintext 'SOLVED' using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$.

Ans: $\det(K) = 3(11) - 8(7) = 33 - 56 = -23$

$$26: -23 + 26 = 3$$

$$\det(K) \equiv 3 \pmod{26}$$

$$\gcd(3, 26) = 1 \quad \checkmark$$

Key is valid.

SOLVED $\rightarrow$ S O L V E D

S O L V E D

18 11 11 21 4 3

$$\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 4 & 11 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} \pmod{26}$$

$$C_1 = (3p_1 + 8p_2) \pmod{26}$$

$$C_2 = (4p_1 + 11p_2) \pmod{26}$$

'SO', $p_1 = 18$ $p_2 = 14$

$$C_1 = 3(18) + 8(14) = 54 + 112 = 166$$

$$166 \pmod{26} : 166 - 6(26) = 166 - 156 = 10 \rightarrow \underline{\underline{K}}$$

$$C_2 = 4(18) + 11(14) = 72 + 154 = 226$$

$$226 \pmod{26} : 226 - 8(26) = 226 - 208 = 18 \rightarrow \underline{\underline{U}}$$

$$SO \Rightarrow \underline{\underline{KU}}$$

'LV', $p_1 = 11$ $p_2 = 21$

$$C_1 = 3(11) + 8(21) = 33 + 168 = 201$$

$$201 \pmod{26} : 201 - 7(26) = 201 - 182 = 19 \rightarrow \underline{\underline{T}}$$

$$C_2 = 4(11) + 11(21) = 44 + 231 = 275$$

$$275 \pmod{26} : 275 - 10(26) = 275 - 260 = 15 \rightarrow \underline{\underline{W}}$$

$$LV \Rightarrow \underline{\underline{TW}}$$

'ED', $p_1 = 4$ $p_2 = 3$

$$C_1 = 3(4) + 8(3) = 12 + 24 = 36$$

$$36 \pmod{26} = 36 - 26 = 10 \rightarrow \underline{\underline{K}}$$

$$C_2 = 4(4) + 11(3) = 16 + 33 = 49$$

$$49 \pmod{26} = 49 - 2(26) = 49 - 52 = -3 \rightarrow 23 \rightarrow \underline{\underline{J}}$$

$$ED \Rightarrow \underline{\underline{KJ}}$$

$$\therefore \text{'SOLVED'} \Rightarrow \text{'KUTW KJ'}$$